Identifying Rocks



Mineral are pure substances that occur naturally and there are about 3000 known minerals in the earth. Minerals can be found on their own and also mixed together in rocks. A mineral is the same substance all the way through.

Many geologists know what the main minerals look like but often need to classify unusual minerals. Keys are used to identify minerals, including ores and minerals found in igneous rocks.

COLOUR and LUSTRE

- Colour refers to the main colour you can see. Many minerals have a range of colours. Sometimes impurities introduce new colours.
- Lustre is the shininess of the surface. Is it metallic (like a shiny new coin) or is it non-metallic (dull)?

HARDNESS

- Hardness is how readily the mineral is scratched. Does your fingernail scratch the mineral? The Mohs scale measures the degree of hardness of different substances. On the Mohs scale, substances are rated from 1 (very soft) to 10 (very hard).

STREAK and CLEAVAGE

- Streak is the colour of the powdered mineral. You see this when you use the mineral to write on something hard, such as the back of a ceramic tile.
- Cleavage is the ability to break along smooth planes, forming flat layers, sharp corners, boxes or other shapes.

Give the meaning of the following words in the paragraphs above:

1. Pure _____

2. Naturally _____

3. Unusual _____

4. Identify _____

5. Refers _____

6. Impurities _____
7. Metallic _____

8. Readily _____

9. Planes _____



Which words or phrases can be replaced with the following:

1. Materials _____

2. Identified _____

3. Combined _____

4. Sort _____

5. Name _____

6. Indicates _____

7. Variety _____

8. Easily _____

9. Scale _____

10. Crushed _____

11. Making _____



<u>Questions</u>

1. What are minerals? _____

   _____

2. How many minerals are known on Earth? _____

   _____

3. Describe what minerals are like? _____

_____

4. Why might a geologist need a minerals key? _____

_____

5. What can introduce new colours to a mineral? _____

6. Why are colours not very useful in identifying minerals? _____

_____

7. What is lustre? _____

8. What does non-metallic mean? _____

9. What does the Mohs scale do? _____

10. What would a 5 on the Mohs scale be like? _____

_____

11. What is streak? _____

12. What is the ability to break along smooth planes called? _____

